

Laboratorio de Microeconomía I

Centro de Investigación y Docencia Económicas

Maestría en Economía - 2025

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Tarea 5

Exercises

Ex. 1.25 A consumer with convex, monotonic preferences consumes non-negative amounts of x_1 and x_2 .

- (a) If $u(x_1, x_2) = x_1^\alpha x_2^{(1/2)-\alpha}$ represents those preferences, what restrictions must there be on the value of parameter α ? Explain.
 - (b) Given those restrictions, calculate the Marshallian demand functions.
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Ex. 1.26 A consumer of two goods faces positive prices and has a positive income. His utility function is

$$u(x_1, x_2) = x_1.$$

Derive the Marshallian demand functions.

Ex. 1.27 A consumer of two goods faces positive prices and has a positive income. His utility function is

$$u(x_1, x_2) = \max[ax_1, ax_2] + \min[x_1, x_2], \quad \text{where } 0 < a < 1.$$

Derive the Marshallian demand functions.

Ex. 3.D.4 Let $(-\infty, \infty) \times \mathbb{R}_+^{L-1}$ denote the consumption set, and assume that preferences are strictly convex and quasilinear. Normalize $p_1 = 1$.

- (a) Show that the Walrasian demand functions for goods $2, \dots, L$ are independent of wealth. What does this imply about the wealth effect (see Section 2.E, Mas-Colell et al. pp. 24-25) of demand for good 1?
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Ex. 3.D.5 Consider again the CES utility function of Exercise 3.C.6, and assume that $\alpha_1 = \alpha_2 = 1$.

- (a) Compute the Walrasian demand and indirect utility functions for this utility function.
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